

Personal project report

グランボット (Grandbot)

a Japanese home companion for the elderly

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October 2026

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1 Introduction

Japan is what we call a super-aged society: more than one in four residents are aged 65 or over, and a large share of later life is spent at home rather than in a clinic (Statistics Bureau, 2025). In this context, ordinary day care and emergency handling matters a lot.

I built "Granbot" for my grandmother. She lives alone in Japan, and although she leads a healthy life, she and the family find comfort in knowing that there exist ways for her to be taking care of daily.

There are two sides to this: her side as an interface constantly on on a dedicated iPad in the living room, and the family side that allows us to monitor things from our phones. Her needs, as this project treats them, are practical rather than diagnostic; the site is not a device for diagnosis or treatment.

The concerns for the application are thus the following: simple UI, implementing daily physical and mental exercise, and making sure the grandmother is and feels safe at home and when going out.

In this report, we describe "Granbot" as it was built, focusing on usefulness of the application for the elderly and the family.

2 How it works

2.1 What the family sets up

A relative creates an account with a name, an email address, and a password. The first login opens a short setup. We store a display name, an age between 60 and 110, the prefecture used for official feeds, the city used in the spoken weather, and a fall-risk flag. We also store the name of the designated emergency shelter, its address, the place where the emergency bag is kept in the elder's house, and one to three family contacts.

When setup is saved, the family site shows an eight-digit pairing code. It expires after 30 minutes and can be used once. On the iPad, that code is entered together with a four-digit PIN, typed twice. The PIN is the daily key for the iPad, and is shared the elder, but the device cookie that remembers the pairing lasts 400 days so there is no need to repeat the eight-digit code each morning. After pairing, the family can issue a new code, or cut the link. Cutting the link advances a generation counter so the old iPad cookies stop being accepted. The family settings also remind us to turn on Guided Access, so the iPad stays inside this page.

The family interface is threefold. First is a list of notices, grouped by day in Japan time. A notice appears when an exercise session is finished, when a meal is marked as had, when a short request is sent, when the elder leaves or comes back, and when an earthquake, heavy-rain, or heat condition is reported. Quake notices are drawn with a stronger border. The family can mark the list as read. The second tab allows the family to modify the elderly's information, and to run practice emergency scenarios. The third and last tab is information on how the site and exercises are built, they serve as justifiers for design decisions.

2.2 The elder's side

The first concern was the UI. Elderly needs easy and largely displayed choices and explanations in a slow Japanese voice. Personifying the interface with a simple cute cat seemed like an appropriate choice. The following choices were made so that elders would be more keen to accept the technology: the cat on the

screen moves its mouth only while a sentence is being spoken, polite です／ます speech is the tone choice, we use short sentences, and there exists a repeat button in case she did not quite catch the content.

The home screen for the elder has six actions: meals, making calls to the family, physical exercises, mental exercises, going out, and music.

- The meals section allows elders to indicate to their family that they have had breakfast, lunch, and dinner; marking them as "yet to be done" for them not to forget. There also is a text box for the elderly to type in food requests for the family.
- The going out section indicates the weather and a reminder to take a phone with enough battery (used for instance in my family's case to track location). When coming back, the elders indicate their return.
- Calls are ordinary phone and FaceTime links for the stored contacts.
- Music is a fixed list of city-pop recordings, played from short preview files, indicating the song name and artist.
- Physical and mental exercises give instructions for daily action.

Moreover and importantly, while the kiosk is unlocked, the iPad tracks every 12 seconds whether a relevant earthquake report exists. If it does, this starts an emergency scenario procedure.

3 Data management

The aim of the data records is to retain what the next morning and the family feed require, and to drop the rest on a schedule. The store is a single SQLite database, reached through Prisma.

Table 1 groups the records by role. The profile is the stable layer: name, age, prefecture, city, fall-risk flag, shelter name, contacts, and the shelter address and bag location. Operational rows are the changing layer: which steps were marked as finished today, and the notices sent to the family. Session material is kept in signed cookies rather than in a session table. Pairing codes are stored only as a hash, and only until they are used or expire.

Record	What it is for	How long it stays
Profile and contacts	Choose routine, feeds, and call list	Kept with the household
Shelter address and bag place	After-shaking and rain sentences	Kept, encrypted
Exercise progress and meals	Show today's buttons and notice for family	Deleted after 30 days
Notices	Family home, grouped by day	Deleted after 30 days
PIN, password, pairing code	Open a session	Hashed; the pairing code is single-use

Table 1: Records kept by Granbot

We separate three operations. Extraction is the reading of an official feed: an earthquake list entry, a warning code, a forecast line. Reconstruction is the filling of a script, which puts stored household words into a sentence whose shape is already fixed. Enrichment is anything the system adds beyond that (ex: clothes to wear, instructions to go to a shelter).

4 Security

This section states the attacks the prototype is built to resist, and the controls that correspond to them. The aim is a household threat model, not a claim of medical-device assurance.

Family pages and family APIs only accept the family cookie, and unlocked kiosk APIs only accept the kiosk cookie. Pairing, PIN unlock, and a public status check are the exceptions on the iPad side. Registration and login are the exceptions on the family side. Cookies are HTTP-only and SameSite; in production they are sent only over HTTPS. Passwords and the PIN are hashed with Argon2id (RFC 9106, 2021). The parameters are a 16-byte salt, three iterations, four lanes, and 64 MiB of memory.

The PIN path has its own limit. Five wrong entries lock that iPad for ten minutes. A separate in-memory limit slows repeated attempts from one address. Pairing is limited in the same process memory, though these limits clear if the server restarts. The pairing code is eight digits, hashed with SHA-256 and a pepper taken from the session secret, and it is cleared when used. Unlinking the iPad increments the generation stored in the device cookie, so a stolen browser profile from before the unlink no longer matches.

The shelter address and the bag location use AES-256-GCM with a fresh 12-byte nonce (NIST, 2007), with the key being a 32-byte value supplied as DATA_KEY. Losing the key makes those two fields unreadable. The session secret signs the cookies with HMAC-SHA256 and must be at least 32 characters.

At this stage, the profile, the scripts, and the sessions have a defined boundary.

5 Physical exercises

5.1 What the sessions contain

Each day has five steps, and the set changes with the weekday. The track itself depends on the profile: below age 75, and without the fall-risk flag, we offer a standing routine done while holding a chair, at 75 or older, or when the family sets the flag, it offers a chair routine. Age is a basically a proxy.

The actions are divided in five categories: marching in place, heel or toe raises, supported balance, sit-to-stand or a partial stand, and easy arm or shoulder movements. The last day of the week is meant to be shorter.

Indicating pain, dizziness, and chest pain stop the session and welcomes the elderly to sit, rest, call the family or 119 if emergency (chest hurting, head spinning, strong pain). Elders can also resume the session at a later time, and completion of the entire set sends a notification to the family.

5.2 Sources

Three public sources justify the design of the physical exercises.

- The World Health Organization asks older adults to include, on three or more days a week, multi-component activity that emphasizes functional balance and strength at moderate or greater intensity, as part of their weekly physical activity (WHO, 2020; Bull et al., 2020).
- The Japanese physical-activity guide for 2023 makes a related recommendation for older adults: walking or an equivalent amount of activity on most days, and multicomponent exercise on three or more days a week. It names gymnastics, dance, and radio calisthenics as familiar examples of

varied movement (MHLW, 2023). Granbot's exercises are meant to be domestic versions of balance and lower-limb strength. The radio-calisthenics day keeps the supported arm pattern and drops the jumps (because jumps do not really match exercises with a chair).

- In a Cochrane review of community exercise, programmes reduced the rate of falls by 23% relative to control (rate ratio 0.77, 95% confidence interval 0.71 to 0.83), and the programmes with the clearest evidence emphasized balance and function (Sherrington et al., 2019). Sherrington et al. (2019) also note that at least one third of community-dwelling people over 65 fall in a year, which is why we kept this all in mind.

Health conditions of course differ from person to person. This design was tailored to match my grandmother's needs, but are highly adaptable.

6 Mental exercises

This section describes the thinking prompts. The aim is to give the day a short orientation and a short memory practice, and to keep that practice from looking like a dementia test or a commercial brain-training product.

6.1 What the sessions contain

The aim of these exercises is to provide the elder with a short orientation and memory practice, and to keep that practice from looking like a dementia test or a commercial brain-training product.

There are seven sets, one for each day, and each set has five prompts.

- Orientation: naming date, weekday, season, part of the day, municipality, weather, year.
- Word repetition: three ordinary words spoken for immediate repetition.
- Reminiscence: childhood breakfast, road to school, a song, festival food, a first job or a household task, a neighbor's face, a trip.
- Categories: naming three things from an ordinary category.
- Prospective cue: drinking a glass of water tonight, washing hands before dinner, opening the curtains tomorrow.

Every exercise mentions that a wrong or missing answer is acceptable, the overarching goal simply being to exercise the mind; the prompts are not scored. Similarly to physical exercises, partial progress is saved and completion notifies the family.

6.2 Sources

Cognitive stimulation, as developed by Spector et al. (2003) and reviewed by Woods et al. (2023), is a structured group activity for people with dementia. Sessions use orientation, reminiscence, word categories, and related discussion, usually for a sustained period and on a schedule. The Cochrane review reports a small benefit on cognitive measures at the end of treatment in that population. Reminiscence work, reviewed by Woods et al. (2018), asks people to talk about the past. These exercises do not prevent dementia, and serve a light cognitive stimulation role.

Again, health conditions differ from person to person. This design was tailored to match my grandmother's needs, but are highly adaptable.

7 Emergency scenarios

7.1 What happens on the iPad

While the kiosk is unlocked, the browser polls a server every 12 seconds, which reads the Japan Meteorological Agency earthquake list. A report is relevant when: the event is no more than 20 minutes old, the prefecture code of the household appears in the intensity list, and the intensity in that prefecture is 4 or higher. A missed feed does not invent an earthquake. When the conditions are met, the screen gets red and a soft siren runs in the browser. The elder is told to hide under a sturdy table, not go outside, and not try to put a fire out. After 20 seconds, or if earthquake stoppage is indicated by the elder, information regarding household emergency bag, shelter, and address are given. Drills mention that they are in fact drills.

Rain is a different official object. Heavy-rain warnings, flood warnings, special heavy-rain warnings, and landslide warning text are treated as an evacuation-level message. The elder is told to take an umbrella and, if necessary, to go to a designated emergency shelter.

Heat uses a shorter rule. A heat advisory or a forecast high of at least 28 °C adds a line about carrying a water bottle, a small fan, and being in a cool room. A low of 12 °C or below, or a high of 16 °C or below, suggests a sweater. These clothing lines are local rules. They are not a clinical heat-illness protocol.

7.2 Sources

Immediate protection during shaking follows public guidelines (JMA, 2024). Intensity 4 is the step at which most people are startled and unsecured objects may move (JMA, 2024b). Our application might indicate emergency scenarios with a delay, and is therefore not always a reliable reaction system, this whole feature is simply for the elder to feel helped and safe.

The rain scenario follows the Cabinet Office distinction of alert level 3, 高齢者等避難: older adults and others who need longer should leave dangerous places for a designated emergency site, earlier than the general order (Cabinet Office, 2021).

8 Limitations

The main limitation is that Granbot was designed for one person and tested in only one household, so its usefulness cannot yet be generalized to a wider elderly population. The exercise routines are also based partly on age and a simple fall-risk flag, rather than on a detailed individual health assessment. Finally, the emergency system depends on external feeds and periodic checks, meaning that alerts may be delayed and should not be considered a fully reliable emergency-warning system.

9 Future work

Listed below are features we could add:

- Daily physical condition assessment with a report to the family in mind.
- Voice control.
- Conversational chat bot, against loneliness issues and for mental stimuli.
- More accurate and detailed emergency reactions.
- Ability to order meals through delivery services, with an interface made more accessible to the elderly.
- Scalability for a wider network of users.

10 Conclusion

Granbot shows how a simple home interface can combine daily routines, physical and mental exercises, family communication, and emergency guidance in one place. The project was designed around the needs of one elderly user and public health recommendation sources, making it possible to focus closely on usability rather than on creating a general-purpose medical system. An important aspect of the project is that it does not try to replace professional care, but instead acts as a small daily companion that can help the elder remain active, connected to the family, and reassured at home.

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